Joint Risk and Review-Budget Certification for Consent-Constrained Incident Alerts

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Abstract

Clinical monitoring must reconcile timely event detection, patient consent and finite review capacity. We show that allowing only the first threshold crossing can make incident missed-event loss non-monotone: a lower threshold may consume the review before the valid warning window. We characterize the minimum feasible review budget using pre-window and in-window score maxima, and apply simultaneous binomial bounds to distinguish certified feasibility, grid infeasibility and insufficient evidence. We also state when consent transformations preserve the sampling assumptions and give a counterexample under consent-mechanism shift. An evaluation on all 40,336 public PhysioNet 2019 patients compares tuned logistic regression, gradient boosting and a GRU across three seeds and two hospitals. Mean pre-onset AUROC reaches 0.794 at hospital A and 0.752 externally. Yet every locally calibrated first-alert grid is excluded at a 5% conditional missed-event target, even with one review per patient. Positive challenge utility does not resolve this failure because it rewards a different warning window and persistent prediction state. Patient-specific consent and reviewer simulations extend the evaluation but do not establish clinical effectiveness. The study identifies a policy-level obstacle that better score discrimination alone does not remove.

1 Introduction

Clinical monitoring systems do more than assign a risk score. They determine which observations are processed, when an alert appears, who receives it and whether someone acts before an opportunity to intervene has passed. Each decision changes the practical meaning of model performance. A score can rank patients well while producing an unmanageable stream of reviews. An alert can receive a favorable benchmark reward yet arrive outside the time window used to define a successful human assessment. A consent restriction can remove an input without removing information already cached in a feature summary or recurrent state.

These distinctions motivate our research question: can a healthcare AI monitoring pipeline remain useful while sensitive data flows, model errors and consequential decisions are governed by explicit consent and effective human oversight? We treat this as a conjunction of requirements. Predictive discrimination, timely event detection, review workload, permitted information flow and correct human action are separate quantities. Evidence for one cannot silently stand in for another.

Sepsis monitoring provides a useful test case because repeated clinical observations, rare events and time-sensitive alerts make these requirements interact. The PhysioNet/CinC 2019 challenge supplied a common benchmark and an explicitly time-dependent utility function [8]. External validation and implementation studies show why evaluation must extend beyond a development-set discrimination score: model performance can deteriorate in another setting, and clinicians' engagement with alerts is itself an empirical question [7, 9].

We investigate a simple operational proposal: reduce repeated alerts by requesting at most one review per patient. The proposal appears attractive because threshold crossings remain nested as the threshold decreases. The relevant clinical loss, however, need not be nested. If the first alert consumes the only review before the valid warning window, lowering the threshold can convert a timely review into a missed incident. This distinction invalidates an unqualified application of monotone-loss conformal risk control to timely first-alert loss.

Our methodological contribution is a precise treatment of that policy mismatch. We characterize population feasibility through pre-window and in-window score maxima, then apply simultaneous binomial confidence bounds to a validation-frozen policy grid. The output has three meanings: a policy is certified to meet both targets, every policy on the grid is excluded, or the sample is insufficient to decide. The confidence machinery follows existing risk-control work; the contribution lies in matching it to the incident-review endpoint and making the operational assumptions explicit.

We evaluate the construction on all 40,336 public PhysioNet patients using tuned logistic regression, histogram gradient boosting and a unidirectional GRU. Patient-level splits separate training, validation, calibration and testing. Additional analyses examine simulated consent changes, hospital-specific recalibration, overlapping demographic groups, noisy review queues and structural review-path checks. The strongest models improve discrimination, but all locally calibrated first-alert grids fail the declared 5% conditional missed-event target. The resulting negative answer is specific: this model and policy family does not demonstrate the conjunction in our research question. It does not establish that a different monitoring or continuing-

review policy cannot do so.

2 Related work

2.1 Clinical prediction, alert utility and deployment

The PhysioNet challenge was designed to make early sepsis prediction comparable across cohorts, clinical variables and evaluation procedures [8]. Its utility rewards prediction at different times relative to onset and penalizes false alarms. That makes it more informative than a single threshold-free ranking metric, but it still evaluates a prescribed prediction representation. Our once-per-patient review count and strict first-review window are additional estimands, not replacements for the challenge's original definition.

Clinical time-series benchmarks have established the value of comparing linear and recurrent models within a common preprocessing and evaluation framework [10]. Missingness is also informative: the fact that a measurement was ordered can predict outcomes, independently of the recorded value. GRU-D explicitly represents missingness and time intervals [11]. Our GRU is a simpler baseline using causal summaries and observation counts; it does not implement learned decay or reproduce GRU-D's architecture. This distinction matters when interpreting the modest model search as evidence about a policy family rather than a comprehensive search for the best sepsis predictor.

Discrimination, probability quality and clinical usefulness also differ. Precision–recall analysis is particularly informative when positives are rare [12]; calibration studies emphasize that apparently strong ranking does not ensure reliable risk probabilities [13]. We therefore tune using validation average precision and report AUROC, average precision, Brier score and challenge utility. Class-weighted training can improve the ordering of cases while changing the scale of the predicted probabilities, so we do not interpret the raw weighted scores as calibrated clinical risks.

The external validation of the Epic Sepsis Model examined both predictive performance and the burden of alternative alerting strategies [9]. The TREWS adoption study examined whether clinicians evaluated alerts and how their interactions related to treatment timing [7]. These studies motivate our separation between an alert being emitted and a correct action being completed. They do not supply the active service-time distribution for our simulated reviewers; response delay includes workflow and availability as well as time spent on the assessment.

2.2 Risk control and constrained policy selection

Conformal risk control bounds an expected loss under exchangeability and the required regularity of a nested loss family [2]. Risk-controlling prediction sets and Learn then Test instead provide high-probability statements about candidate procedures, with selection justified by valid tests or simultaneous confidence statements [3, 4]. Pareto Testing addresses the efficient selection of configurations subject to multiple risk constraints [5]. Our finite-grid construction is deliberately elementary: exact binomial intervals and a union bound provide a transparent reference method for a non-monotone first-alert loss.

The distinction between the guarantees is consequential. An expectation over calibration and a future patient is not the same object as a high-probability upper bound on the population risk of a fitted rule. Likewise, failing to pass a test is not evidence that every candidate is infeasible. We use lower bounds to support the latter statement and retain an undetermined outcome when neither direction is established. We make no claim that the underlying binomial, union-bound or importance-weight arguments are new.

Recent clinical selective-prediction work combines conformal prediction with cost-aware deferral and evaluates group-conditional and shifted settings [1]. Such work is closely related in motivation. Our loss is a missed incident among event-positive patients, and our capacity variable is the number of patient reviews. Prediction-set coverage, error on retained cases and timely first review are different random variables, so a guarantee for one does not transfer automatically to the others. Under hospital shift, weighted conformal methods further require a relationship between source and target distributions [22]; an estimated domain-classifier ratio alone does not verify that relationship.

2.3 Consent and meaningful oversight

Dynamic consent treats permission as an ongoing interaction in which participants can manage preferences as data use evolves [14]. We model one technical consequence of changing permissions: which inputs and derived state are available to a predictor. This supports controlled sensitivity experiments but omits comprehension, communication, preference formation and the institutional process through which consent is obtained.

Meaningful human control has also been formulated in terms of systems responding to relevant human reasons and outcomes remaining attributable to responsible people [15]. A review node in an information graph captures only a routing requirement. It says neither that a reviewer has enough information nor that the reviewer has time, competence or authority to act. Our graph checks and stochastic reviewer model address distinct parts of that larger process, while prospective human evaluation remains outside the present study.

3 Study design and data

3.1 Operational questions and endpoints

The primary analysis asks whether a frozen predictor and threshold policy can meet two declared constraints simul-

Table 1: Public cohort and incident eligibility. "Positive" means at least one positive label. The two exclusion categories concern the strict incident endpoint; all patients remain in workload calculations.

Quantity	Hospital A	Hospital B
Patients	20,336	20,000
Hourly records	790,215	761,995
Label-positive patients	1,790	1,142
Initial-positive patients	203	223
Onset at/after observed end	2	13
Eligible incident patients	1,585	906

taneously: conditional incident missed-event risk at most $\alpha = 0.05$ and expected review workload at most K. We examine $K \in \{0.1, 0.25, 0.5, 0.75, 1\}$ reviews per patient. These values define a sensitivity analysis over review budgets; they are not estimates of capacity at a particular hospital. The statistical confidence parameter is $\delta = 0.05$ within each model and frozen policy grid.

We also evaluate benchmark usefulness using positive normalized challenge utility. This is an operational comparison with the challenge's inaction reference, not a clinical benefit threshold. The full research question additionally requires permitted data flow and effective review. Structural checks assess the specified consent and review graph, while the queue experiment measures timely and correct action only within a declared simulation. We do not combine these evidence types into a single clinical-safety score.

3.2 Public cohort and provenance

We use version 1.0.0 of the public PhysioNet/CinC 2019 release [6, 8]. Hospital A contributes 20,336 patient files and 790,215 hourly records; hospital B contributes 20,000 files and 761,995 records. Together they contain 1,552,210 rows. Each record contains 40 input variables: eight vital signs, 26 laboratory variables and six demographic or administrative variables, plus the supplied SepsisLabel. No video, skeleton or credentialed dataset contributes to the revised primary analysis.

The cohort is defined by the full public-file manifest, not by selecting a favorable subset. Downloads are checked against the published object ETag, and a SHA256 digest is recorded for each local file. Patient IDs, site membership, row boundaries and split assignments are retained in the released provenance. The public source is already deidentified; the study does not link records to external identities or collect new patient information. Dataset terms and evaluator licensing accompany the release.

3.3 Onset reconstruction and monitoring interval

The supplied label becomes positive six hours before inferred sepsis onset. Let a_i be the index of patient i's first positive label and set $o_i = a_i + 6$. We use zero-based hourly indices. The strict warning window is $[a_i, o_i)$, containing the six observations immediately before inferred onset. The label is not shifted a second time.

A patient enters the conditional incident-risk denominator when $a_i > 0$ and $o_i < T_i$, where T_i is the number of observed rows. Initial-positive patients have no observed pre-window prefix, and patients whose inferred onset is at or beyond the observed end do not meet this complete-window definition. Table 1 reports these exclusions. This retrospective convention is stricter than simply asking whether a positive label ever occurs.

For workload, positive patients are monitored through $\min(o_i, T_i)$ and patients without a positive label through T_i. All patients, including those excluded from incident risk, remain in the workload denominator. The stopping time depends on the retrospective label, which would not be known to a deployed model. It therefore defines a benchmark accounting interval, not a prospective operational stopping mechanism. Full-stay challenge utility is computed separately. A later deployment study would need an observable endpoint, such as discharge or an independently recorded clinical decision, and would have to reevaluate workload under that endpoint.

3.4 Patient-level splits

For each seed in $\{11, 29, 47\}$, a seeded permutation of hospital A assigns 55% to training, 15% to validation, 15% to calibration and 15% to testing. Integer cut points produce 11,184 training, 3,051 validation, 3,050 calibration and 3,051 test patients. Hospital B is independently permuted into 5,000 local-calibration and 15,000 external-test patients. No B labels train the clinical predictor.

Every observation from a patient stays in one partition within a seed. Hyperparameters and candidate thresholds use training and validation only. Calibration is reserved for risk/workload certification; test data evaluate the frozen result. The three seed-specific test sets overlap, so the analysis reports seed variability descriptively rather than treating the runs as three independent clinical cohorts. Splitting is not stratified on the outcome; the positive calibration counts are reported explicitly.

3.5 Causal representation and missing data

At hour t, each of the 40 raw variables contributes its most recent observed value up to t. For the 34 physiological variables, we additionally calculate the expanding mean of observed values and the cumulative observation count. The feature vector therefore has 40 + 34 + 34 = 108 entries. If x_{itv} is variable v and m_{itv} indicates observation, the summaries are

$$c_{itv} = \sum_{u \le t} m_{iuv}, \qquad \bar{x}_{itv} = \frac{\sum_{u \le t} m_{iuv} x_{iuv}}{c_{itv}}, \tag{1}$$

$$x^{\text{last}}_{itv} = x_{iu^*v}, \quad u^* = \max\{u \le t : m_{iuv} = 1\}. \tag{2}$$

The mean and last value remain missing until an observation exists; the count is initially zero. SepsisLabel is excluded from the input. Forward filling and expanding summaries use only the current prefix. The code does not interpolate backward from a later measurement. Tests perturb future observations and verify that earlier feature values do not change.

Observation counts intentionally retain information about the measurement process. This can improve prediction but also makes consent interventions consequential: masking a value while retaining its historical count would leave a prohibited information path. Missingness-aware clinical models motivate treating the observation process explicitly [11]; our count representation remains simpler than a dedicated missingness-decay architecture.

4 Prediction models and training

All models predict the hourly supplied label using the same causal representation. Training loss and the primary discrimination evaluation omit rows from inferred onset onward. For a positive patient, the final included row is $\min(o_i, T_i) - 1$; all observed rows of a label-negative patient are eligible. This prevents long post-onset stays from dominating the primary comparison. It does not remove the six pre-onset positive labels that define the prediction target.

4.1 Logistic regression

The linear baseline imputes missing features with training-set means, retains initially empty feature columns, and applies training-set standardization. It fits an L2-regularized logistic model

$$s_{it} = \sigma(\beta^\top z_{it} + b), \tag{3}$$

where z_{it} is the transformed feature vector. Four candidates combine inverse regularization $C \in \{0.1, 1\}$ with either unweighted or class-balanced loss. The iteration limit is 350 and the random state follows the split seed. The selected model maximizes hourly validation average precision. This is a tuned linear baseline; it is not the fixed-C predictor used in the archived initial study.

4.2 Histogram gradient boosting

The nonlinear tabular baseline uses histogram gradient boosting, building an additive tree score under binary classification loss [19]. Training-only variance screening removes constant and entirely missing columns; remaining missing values are handled by the estimator. The screening is necessary for the recorded implementation and uses no validation, calibration or test information.

The three candidates are $(15, 1, \text{unweighted})$, $(31, 5, \text{unweighted})$ and $(31, 5, \text{balanced})$, where the entries are maximum leaf count, L2 regularization and class weighting. Every candidate uses 120 boosting iterations, learning rate 0.08 and disabled automatic early stopping. Validation average precision chooses the candidate. The implementation uses scikit-learn [23]; the locked software version and exact parameter dictionaries are released. This small grid tests a stronger nonlinear predictor without claiming exhaustive hyperparameter optimization.

4.3 Unidirectional recurrent model

The sequence baseline has one unidirectional GRU layer [20] and a linear logit head at each hour:

$$h_{it} = \text{GRU}(z_{it}, h_{i,t-1}), \qquad s_{it} = \sigma(w^\top h_{it} + b). \tag{4}$$

Patient sequences are padded within batches and packed by their true lengths. Padding contributes no loss. The recurrent state is reset between patients. Although complete observed sequences can be passed through the forward computation, post-onset training positions are masked out of the loss; unidirectionality prevents later observations from influencing earlier outputs.

Training-set means and standard deviations normalize the 108 features. Empty or near-constant scales are replaced by one; normalized missing entries become zero and all normalized values are clipped to $[-10, 10]$. Candidate hidden sizes are 32 and 64. Each candidate receives six epochs with Adam [21], learning rate 0.001, batch size 64 and gradient-norm clipping at one. Training patients are reshuffled each epoch using the split seed.

The positive-weighted binary cross-entropy is

$$\mathcal{L} = -\frac{1}{|\mathcal{I}|} \sum_{(i,t)\in\mathcal{I}} \left[\omega y_{it} \log s_{it} + (1 - y_{it}) \log(1 - s_{it})\right], \tag{5}$$

where $\mathcal{I}$ contains valid pre-onset training positions and $\omega = \min\{20, (1 - \bar{y})/\bar{y}\}$ uses the training prevalence. The implementation averages this loss over valid positions in each batch. Every epoch is evaluated on validation average precision; the best hidden-size/epoch pair is saved. There is no calibration-set early stopping, test-set model selection or bidirectional recurrence.

4.4 Selection, frozen scores and search scope

For all three model families, validation average precision is the sole model-selection criterion. AUROC is logged for the tabular candidates as a diagnostic. The selected estimators are frozen before calibration and before any consent or queue stress test. Prediction arrays preserve patient order and row boundaries for validation, A calibration, B calibration and both test sites. This allows later policy audits to reuse scores without retraining.

The search budgets differ by family: four logistic fits, three boosting fits and two six-epoch GRU trajectories per seed. They are appropriate for comparing these implementations, but they do not establish an architecture-level ranking under equal compute or equal tuning effort. We report both the settings and the selection rule so that a stronger

model search can extend the study without changing its endpoints.

5 Alert policies and joint guarantees

5.1 Patient-level estimands

A patient supplies a finite sequence of causally available observations, consent states, and frozen scores $S_1, \dots, S_T \in [0, 1]$. Let Z indicate an incident event with an observed valid warning window $[a, b]$. For a threshold τ, the first-alert policy is

$$\begin{aligned} T_\tau &= \inf\{t : S_t \geq \tau\}, \quad R_\tau = \mathbf{1}\{T_\tau \leq T\}, \\ L_\tau &= \mathbf{1}\{T_\tau \notin [a, b]\}, \quad Z = 1. \end{aligned} \tag{6}$$

An empty set gives $T_\tau = \infty$. Assume $0 < \Pr(Z = 1) < 1$ and $0 < \alpha < 1$. We target conditional incident risk $r(\tau) = \mathbb{E}[L_\tau \mid Z = 1]$ and expected workload $w(\tau) = \mathbb{E}[R_\tau]$, measured in reviews per patient. Their joint constraints are $r(\tau) \leq \alpha$ and $w(\tau) \leq K$. For m independent patient arrivals in a period, the expected number of reviews is $mw(\tau)$; an expectation bound does not guarantee a finite waiting time.

For comparison, a repeated-alert policy uses every threshold crossing. Its incident missed-event loss is $\mathbf{1}\{\max_{a \leq t \leq b} S_t < \tau\}$, which is monotone in τ. A rule that emits only the first crossing has lower review count but a different risk.

Proposition 1 (First-alert non-monotonicity)**.** *Nested threshold crossings do not imply monotone timely first-alert loss. There is a trajectory for which lowering τ changes the incident loss from zero to one, although the probability of any review is non-decreasing as the threshold decreases.*

Proof. Let a pre-window score equal 0.6, an in-window score equal 0.9, and all other scores equal zero. At threshold 0.8, the first alert lies inside the window and loss is zero. At threshold 0.5, the first alert occurs before the window and consumes the only alert; loss is one. Both thresholds trigger exactly one review. A threshold above 0.9 yields no alert and loss one. Thus the loss sequence as thresholds increase is $1, 0, 1$. Any-review events remain nested because they depend only on the overall score maximum. □

This prevents applying the monotone-loss conformal risk-control theorem [2] to the first-alert loss without additional assumptions. Counting any in-window crossing while reporting only one early review would evaluate a different policy.

5.2 Population feasibility

Define the pre-window maximum $A = \max_{t<a} S_t$, with $A = -\infty$ for an empty prefix, the in-window maximum $B = \max_{a \leq t \leq b} S_t$, and the maximum C over the monitoring interval used to count reviews. For event-positive patients the monitoring interval includes the warning window. Let $\pi = \mathbb{P}(Z = 1)$ and

$$\begin{aligned} H(\tau) &= \mathbb{P}(A < \tau \leq B \mid Z = 1), \\ W(\tau) &= \pi\mathbb{P}(C \geq \tau \mid Z = 1) \\ &\quad + (1 - \pi)\mathbb{P}(C \geq \tau \mid Z = 0). \end{aligned} \tag{7}$$

Theorem 2 (Exact feasibility of a first-alert threshold family)**.** *For a fixed threshold domain $\mathcal{T}$, a jointly feasible first-alert threshold exists if and only if*

$$\{\tau \in \mathcal{T} : H(\tau) \geq 1 - \alpha,\ W(\tau) \leq K\} \neq \varnothing.$$

If $\mathcal{T}_\alpha = \{\tau \in \mathcal{T} : H(\tau) \geq 1 - \alpha\}$ is nonempty and has a maximum $\tau_\star$, the minimum feasible expected review budget is

$$K_{\min} = W(\tau_\star). \tag{8}$$

Consequently, joint feasibility is equivalent to $K \geq K_{\min}$. Moreover $K_{\min} \geq \pi(1-\alpha)$. If $\mathcal{T}_\alpha$ is empty, no review budget makes this threshold family feasible.

The maximum assumption always holds for a finite nonempty policy grid. For a continuous domain it is a substantive attainment condition; replacing a minimum by an infimum without attainment can make a boundary feasibility assertion false. This characterization is exact but needs the joint law of A and B, not merely an hourly ROC curve.

Corollary 3 (Closed form when early consumption is absent)**.** *Suppose no alert is possible before the valid window, the positive-patient review interval equals that window, and the conditional distributions of C are continuous, with positive-class CDF F_1 strictly increasing at its unique α-quantile q_α. Let the negative-class CDF be F_0. Then*

$$\begin{aligned} K_{\min} &= \pi(1 - \alpha) + (1 - \pi)\{1 - F_0(q_\alpha)\}, \\ q_\alpha &= F_1^{-1}(\alpha). \end{aligned} \tag{9}$$

This is the familiar threshold/ROC workload identity under an explicit no-early-consumption assumption; it is not a general formula for an unrestricted first-alert policy.

5.3 Finite-sample joint certification

Let $\mathcal{T} = \{\tau_1, \dots, \tau_M\}$ be fixed independently of calibration data, for example using training and validation only. Calibration patients are independent and identically distributed from the deployment population, including the consent mechanism. For each policy compute positive-patient losses and all-patient review indicators. Let n_+ be the positive calibration count and n the total count. Use exact one-sided Clopper–Pearson binomial bounds [16] with tail probability $\eta = \delta/(4M)$ to obtain (ℓ_j^r, u_j^r) and (ℓ_j^w, u_j^w).

Theorem 4 (Joint certificate and honest abstention)**.** *Conditional on any independently trained predictor and frozen grid, with probability at least $1 - \delta$ over calibration, all of the following hold simultaneously:*

1. *Every j with $u_j^r \le \alpha$ and $u_j^w \le K$ satisfies both population constraints.*
2. *If every j has $\ell_j^r > \alpha$ or $\ell_j^w > K$, no policy on this grid satisfies both constraints.*
3. *Any data-dependent selection among certified policies, including the one with smallest workload upper bound, preserves assertion 1.*

No monotonicity of L_τ is required. If neither condition 1 nor condition 2 is met, the output is undetermined, not infeasible. For $n_+ = 0$, the risk interval is $[0, 1]$ and no positive-risk claim is certified.

The construction applies simultaneous-confidence/LTT reasoning to the first-alert policy in Proposition 1. The three outputs distinguish an acceptable operating point, an excluded grid, and a calibration sample that cannot resolve the question.

The certificate is a high-probability statement about the population risk of the selected policy, distinct from the expectation-over-calibration guarantee of ordinary CRC. Model selection performed on calibration would require including every candidate model/policy in the multiplicity family; here model tuning uses validation only. Separate per-model certificates are not automatically simultaneous across models or seeds.

Corollary 5 (Prespecified group or site cells)**.** *For groups $g = 1, \dots, G$ specified independently of calibration, apply Theorem 4 within each cell using δ_g with $\sum_g \delta_g \le \delta$. All certified group-conditional risk and workload statements then hold jointly with probability at least $1 - \delta$. The groups may overlap. Empty or underpowered cells can remain undetermined. A rule calibrated in one hospital is not certified in another merely by relabeling its population.*

5.4 Consent, exchangeability and shift

Let U denote the complete patient trajectory, C the possibly time-varying consent trajectory, and V any policy randomization. A frozen causal procedure $\Phi(U, C, V)$ produces permitted features, scores, alert decisions, and losses. The procedure may clear cached features and recurrent state after revocation.

Proposition 6 (Sufficient exchangeability condition)**.** *If the calibration and future tuples (U, C, V) are identically distributed and independent, and the same frozen measurable transformation Φ is applied to every tuple, their loss and review vectors are identically distributed and independent. Theorem 4 therefore remains valid, including after conditioning on positive-event membership. Consent need not be independent of the outcome.*

The weaker condition $C \perp Y \mid X$ alone does not guarantee transfer between different consent populations. For a counterexample, let $X = Y \sim \text{Bernoulli}(1/2)$ and score $S = CX$. Calibration consent is always $C = 1$, deployment consent always $C = 0$, and the threshold is $1/2$. Both populations satisfy $C \perp Y \mid X$, yet positive-patient missed risk changes from zero to one. Conditional ignorability does not make the joint distributions equal.

Weighted conformal methods address particular distribution shifts when suitable likelihood-ratio information is available [22]. Appendix A gives a bounded-ratio risk/workload certificate derived from Hoeffding's inequality [17]. Its known-ratio assumption is stronger than fitting a hospital classifier. Our empirical weighted comparison is consequently a diagnostic, not a guaranteed hospital-transfer procedure.

6 Consent, calibration and oversight evaluation

6.1 Threshold family and certificate computation

For each frozen predictor, we pool its validation scores and take 51 equally spaced empirical quantiles from 0 to 1, adding thresholds 0 and 1.000001. Duplicate values are removed. The first endpoint alerts whenever monitoring is nonempty, and the last emits no alert for scores in $[0, 1]$. The resulting grid is independent of the calibration and test outcomes conditional on training and validation.

For each calibration patient and threshold, the code records an incident-loss indicator, an any-review indicator and the time of the first crossing. Conditional risk uses only incident-eligible patients; workload uses every calibration patient. Let k_j^r be missed incidents among n_+ positives and k_j^w be reviewed patients among n total patients. For either binomial count (k, N), an exact one-sided interval with tail probability η has endpoints

$$\ell(k, N) = \begin{cases} 0, & k = 0, \\ \text{Beta}^{-1}(\eta; k, N - k + 1), & k > 0, \end{cases} \tag{10}$$

$$u(k, N) = \begin{cases} 1, & k = N, \\ \text{Beta}^{-1}(1 - \eta; k + 1, N - k), & k < N. \end{cases} \tag{11}$$

We set $\eta = \delta/(4M)$ for M grid policies, applying the bounds to both risk and workload [16]. A zero-size sample receives $[0, 1]$. Theorem 4 then justifies the three outputs used throughout the results.

Among certified policies, the implementation selects the smallest workload upper bound. If none is certified, it does not deploy an empirically attractive policy under the name of a certificate. A grid-infeasible result requires every candidate to violate at least one lower-bound test; otherwise the result remains undetermined. Computation uses $O(nM)$ indicator work and storage in the present implementation. Streaming counts could reduce storage, but no such optimization is needed for these cohort sizes.

Finite calibration size is relevant even when no misses are observed. For $k = 0$, the upper risk bound is $1 - \eta^{1/n_+}$.

Algorithm 1 Joint certification on a frozen grid

Require: Patient losses L, reviews R, eligibility Z
Require: Frozen thresholds $\mathcal{T}$, targets (α, K, δ)
$M \leftarrow |\mathcal{T}|$; $\eta \leftarrow \delta/(4M)$
for $j = 1, \ldots, M$ **do**
    Compute exact risk bounds (ℓ_j^r, u_j^r) on $Z = 1$
    Compute review bounds (ℓ_j^w, u_j^w) on all patients
end for
$F \leftarrow \{j : u_j^r \le \alpha \text{ and } u_j^w \le K\}$
if $F \neq \varnothing$ **then**
    return certified, $\arg\min_{j \in F} u_j^w$
else if $\ell_j^r > \alpha$ or $\ell_j^w > K$ for every j **then**
    return grid infeasible
else
    return undetermined
end if

Thus zero observed misses can meet α only when

$$n_+ \ge \left\lceil \frac{\log \eta}{\log(1-\alpha)} \right\rceil. \tag{12}$$

For $M = 53$, $\alpha = \delta = 0.05$, this requires 163 positives. The actual local calibration pools contain 219–253 positives in A and 230–242 in B. Their failure is therefore not explained solely by a correction that would reject even a zero-miss rule.

6.2 Hospital transfer and demographic cells

We distinguish three evaluations: A calibration with A testing, A calibration with B testing, and B calibration with B testing. The first and third target a locally matched patient distribution. A frozen A rule has no asserted B guarantee merely because the same score function is used. B recalibration changes the threshold evidence, not the clinical predictor, which remains trained on A.

Four demographic cells are prespecified: female, male, age below 65 and age at least 65. Sex and age cells overlap. Each receives failure budget $\delta/4$ and is evaluated at $K = 0.5$. The code reports positive calibration support and group-specific certificate status. Sparse groups may remain unresolved even when a population-wide claim is available. The analysis does not infer demographic fairness from an aggregate certificate, and the available binary recorded gender variable does not represent the full range of gender identities.

A supplementary hospital-shift diagnostic fits a logistic domain classifier to the first available feature row of A training patients and B calibration patients. Its estimated B probability $q(x)$ gives a sample-size-adjusted density-ratio estimate, clipped to $[0.05, 20]$. On incident-positive A calibration patients, we compute weighted repeated-window loss and a CRC-inspired correction using the maximum fitted weight. The effective positive sample size is $(\sum_i w_i)^2/\sum_i w_i^2$. The resulting threshold is tested on B. Because the ratio is estimated and clipped and conditional invariance is unverified, this diagnostic is not an implementation of the known-ratio theorem with an exact guarantee.

6.3 Calibration-subset stability

To examine the role of calibration size, we repeatedly sample 25, 50, 100, 200 or 400 incident-positive calibration patients without replacement, when the local pool is large enough. There are 200 draws per available size, model, site and seed. Within a draw, a simultaneous one-sided risk upper bound uses tail probability $\delta/(2M)$. If multiple policies pass, the policy with the lowest empirical review frequency in the all-patient calibration pool is selected. This analysis certifies risk only; it does not add a workload guarantee to the subset experiment.

We record whether a rule is certified, its realized test risk when it is, and whether that risk exceeds α. Draws reuse the same calibration pool and test set and are therefore dependent. Their empirical frequency of exceedance is a stability diagnostic rather than a binomial estimate from independent deployment trials. If no rule is certified, the conditional exceedance frequency is undefined. It must not be reported as zero or interpreted as evidence of perfect coverage.

6.4 Discrimination, utility and uncertainty

Primary hourly discrimination uses only the pre-onset rows defined in Section 3. AUROC measures ranking across positive and negative rows; average precision summarizes the precision–recall curve; Brier score is the mean squared difference between a score and the binary label. Average precision is particularly relevant to rare positive labels [12]. Brier score reflects overall probability accuracy and should not be labeled a pure calibration metric [13].

For every model, site and seed, we draw 500 bootstrap samples of whole patients with replacement. All rows of a sampled patient receive the same multiplicity. AUROC and Brier score are recomputed using those multiplicities, and the 2.5th and 97.5th percentiles form the reported intervals. This cluster-level use of the bootstrap [18] preserves within-patient temporal dependence and treats patients as the sampling units. It conditions on the fitted model and split; it does not include uncertainty from a new model search or a new hospital population.

The normalized challenge utility is

$$U = \frac{U_{\text{observed}} - U_{\text{inaction}}}{U_{\text{optimal}} - U_{\text{inaction}}}, \tag{13}$$

using the official additive utility implementation [8]. For repeated alerts, predicted condition state is the hourly threshold decision. For the first-alert policy, the state remains positive after the first crossing. This latched state represents persistent positive prediction, not a new review at every later hour. Utility is evaluated over the full stay, while

review count and incident loss follow the retrospective monitoring definition. Tests compare the vectorized utility calculation against the pinned official evaluator.

A separate operating-point audit selects the threshold that maximizes validation utility for each model and policy, then freezes it before evaluating either test site. For these fixed policies, exact pointwise 95% binomial intervals describe conditional incident risk. They are distinct from the simultaneous calibration certificate. The audit also decomposes first-alert misses into an early review and no review before onset. Test-frontier points are used only to describe the tradeoff surface, never to choose the reported operating policy.

6.5 Patient-specific consent and revocation

We simulate permission for two input groups, laboratory measurements and vital signs. For each patient, each group is denied independently with probability $p \in \{0, 0.1, 0.3, 0.5\}$. Thus p is a per-group probability, not the proportion of patients with any denial; the latter is $1 - (1 - p)^2$. Demographic and administrative variables remain available. The experiment studies heterogeneity across patients rather than uniformly removing a feature block from the whole population.

In the admission opt-out scenario, denied last values and expanding means are set to missing throughout the trajectory, while denied observation counts are set to zero. This removes both recorded values and the associated measurement-history summaries. The previously fitted imputer or missing-value handling is then applied without retraining. The intervention does not infer values from future observations or preserve a denied historical count.

In the revocation scenario, the same denial decision takes effect at zero-based hour 12. Denied summaries are cleared from that point onward, including information accumulated before withdrawal. For the GRU, the sequence is split at withdrawal and the subsequent segment starts with a new hidden state. This conservative reset also loses permitted information encoded in the previous state; the experiment does not attempt selective deletion from a mixed latent representation.

For each scenario, calibration and both test sites are rescored using the frozen predictor. A calibration set from A is used to seek a joint certificate at $K = 0.5$. When no certificate exists, the descriptive cost analysis uses a labeled empirical fallback: minimum-workload threshold among those with empirical risk at most 0.05, or minimum empirical risk if none qualifies. This fallback has no asserted risk guarantee. Consequently, consent curves combine the score intervention with scenario-specific empirical threshold selection and are not a fixed-threshold causal effect.

Scenario seeds are reused to make permission patterns reproducible and comparisons stable. The stress experiment is not an independent repeated-sampling validation of the consent theorem. That theorem requires the complete patient/consent tuples to obey its stated sampling law. Real preferences, correlations between consent and health status, communication quality, and strategic selection are not observed in this dataset.

6.6 Reviewer queue and end-to-end loss

The queue uses the validation-selected first-alert threshold and the actual held-out patient score trajectories. Patient admission times are generated by exponentially distributed interarrival times with rates 0.5, 1 or 2 patients per hour. Each patient's request occurs at simulated admission plus that patient's first-alert hour. Requests are sorted by arrival time and handled by one first-come, first-served reviewer.

For request j at time a_j, constant service time s and previous completion d_{j-1}, waiting and completion are

$$q_j = \max\{0, d_{j-1} - a_j\}, \qquad d_j = \max\{a_j, d_{j-1}\} + s. \tag{14}$$

Service times are 2, 5 or 10 minutes. Base reviewer sensitivity is 0.8, 0.9 or 1, and specificity is 0.9 or 0.99. A declared fatigue rule multiplies sensitivity by $\max\{0.5, 1 - 0.05q_j\}$ when waiting is measured in hours. The resulting sensitivity governs a Bernoulli correct-review draw. A false consequential action is drawn with probability one minus specificity only for patients whose entire label sequence is negative.

An incident succeeds end to end only if its first alert lies in the valid warning window, review is correct, and completion occurs before inferred onset. A perfect reviewer cannot recover an incident with no timely request under this policy. The simulation reports mean and 95th-percentile waiting, review count, false actions and conditional end-to-end missed risk. The $3 \times 3 \times 3 \times 2$ parameter combinations give 54 scenarios per model, site and seed, or 972 overall.

These service, arrival and fatigue values are sensitivity assumptions. The TREWS study supplies evidence that response and adoption must be measured [7], but response delay is not an estimate of active review duration. The queue therefore does not establish a staffing recommendation, a hospital-specific workload limit or an empirical model of clinician cognition. The corrected audit excludes all label-positive patients from the false-action class, including positives who are ineligible for the strict incident endpoint.

6.7 Information-flow and review-path checks

The structural component represents data sources, transformations, storage, review and consequential actions as a directed graph. Each edge carries the patient and processing purpose to which permission applies. Applying a consent mask removes prohibited flows; withdrawing a source invalidates cached state that depends on it. This model makes the tested policy explicit, but it does not inspect an external hospital application or prove that every real data flow has been represented.

Mandatory review is checked as a path property. For sensitive source set S, consequential destinations D and review

Table 2: Pre-onset discrimination, mean across three seeds. AP denotes average precision. Per-seed 95% patient-bootstrap intervals appear in Table 4; seed averaging is not an independent-cohort meta-analysis.

Model	AUROC A	AUROC B	AP A	AP B
Logistic	0.745	0.666	0.049	0.025
Boosting	0.794	0.731	0.060	0.033
GRU	0.778	0.752	0.060	0.031

nodes H, every S-to-D path intersects H exactly when deleting H disconnects S from D. This standard equivalence is tested by reachability after node removal. A reviewer placed somewhere in a graph is insufficient if a bypass remains.

To move beyond the original small constructed graph, we generate 500 DAGs with 50–300 nodes. For ordered nodes $i < j$, an edge is included independently with probability $3/n$. Approximately 10% of internal nodes are assigned as review nodes; node 0 is the source and node $n - 1$ the action. We report whether an action path exists and whether one survives removal of review nodes. These simulations assess structural behavior across a broader family. They do not estimate the prevalence of bypasses in deployed clinical systems, and they do not turn a path constraint into evidence of effective human review.

7 Results

7.1 Predictive performance

Table 2 summarizes all three splits. Boosting has the highest mean AUROC in A; the GRU has the highest mean AUROC in B. These are descriptive rankings: the seed-specific test populations overlap. Average precision remains low, and the class-weighted models are not calibrated probabilities. Full patient-bootstrap intervals and Brier scores are in the appendix.

7.2 First-alert feasibility

For every model and seed, the local A and B calibrations certify grid infeasibility at $\alpha = 0.05$, for every tested budget from 0.1 to 1 review per patient. There are 90 local calibration/budget outcomes; each certificate retains its stated per-grid confidence interpretation. The result concerns the frozen grids, not all thresholds or all monitoring designs. Increasing review capacity cannot repair a grid whose first-alert timing already violates the risk target.

Figure 1 also shows the realized test risks. Even the best test-grid point, selected retrospectively only to describe the frontier, misses at least 90.0% of eligible incidents. This stringent endpoint treats a review just before the six-hour window as a miss, even if it might be useful clinically. The negative result is therefore about timely *first* review under this definition, not about whether early alerts have clinical value.

7.3 Operating policies and failure mechanisms

We additionally choose thresholds by validation challenge utility and freeze them before testing (Table 3). Boosting in A achieves mean utility 0.332 with 0.193 reviews per patient, yet its strict incident missed risk is 0.960. Its three test splits contain 742 eligible incident observations in total, counting overlap across splits: 414 receive an early first alert and 298 receive none before onset. Only 30 receive their first alert in the specified window. The challenge utility credits a broader time window and a persistent positive state; it is not a surrogate for this first-review endpoint.

Repeated alerts recover more incidents but require several reviews per patient. At validation-selected utility thresholds, no model/site/policy combination has all three of positive utility, risk at most 0.05, and workload at most 0.5. This is the empirical answer to the declared joint target, not proof against alternative cooldown or continuing-review policies.

Figure 2 separates the two mechanisms. An earlier review is not equivalent to an unreviewed patient, but both fail the strict first-review window. This decomposition should accompany any interpretation of the aggregate missed-risk value.

7.4 Recalibration and group support

Local B calibration contains 230–242 eligible positives, compared with 219–253 in A. None of 14,400 overlapping calibration subsets (25, 50, 100 or 200 positives) certifies the 5% first-alert risk target. Thus the fraction of test-risk violations *among certified rules* is undefined, not zero. Four hundred positives are unavailable in these calibration pools; the study does not establish how much additional data would suffice. All 72 prespecified group/model/site/seed checks also exclude their tested grids.

7.5 Consent costs and reviewer simulation

Consent interventions do not restore feasibility. They change both scores and the descriptive fallback threshold, so a reduced review count need not be an improvement. For example, GRU opt-out at 50% per feature group changes mean A missed risk from 0.952 to 0.979 while reviews fall from 0.207 to 0.104 per patient. These are simulated permissions, not observed patient preferences. The estimated-ratio repeated-alert diagnostic has external-site risk from 0.034 to 0.152 across model/seed combinations and carries no exact shift guarantee.

The corrected reviewer audit evaluates 972 queue scenarios on frozen scores. Across all these scenarios, end-to-end missed risk ranges from 0.912 to 0.987; a perfect reviewer cannot recover an incident that never receives a timely review. The simulation is not evidence about real clinician performance. Among 500 random DAGs, 126 contain a source-to-action path and 103 contain a path bypassing the

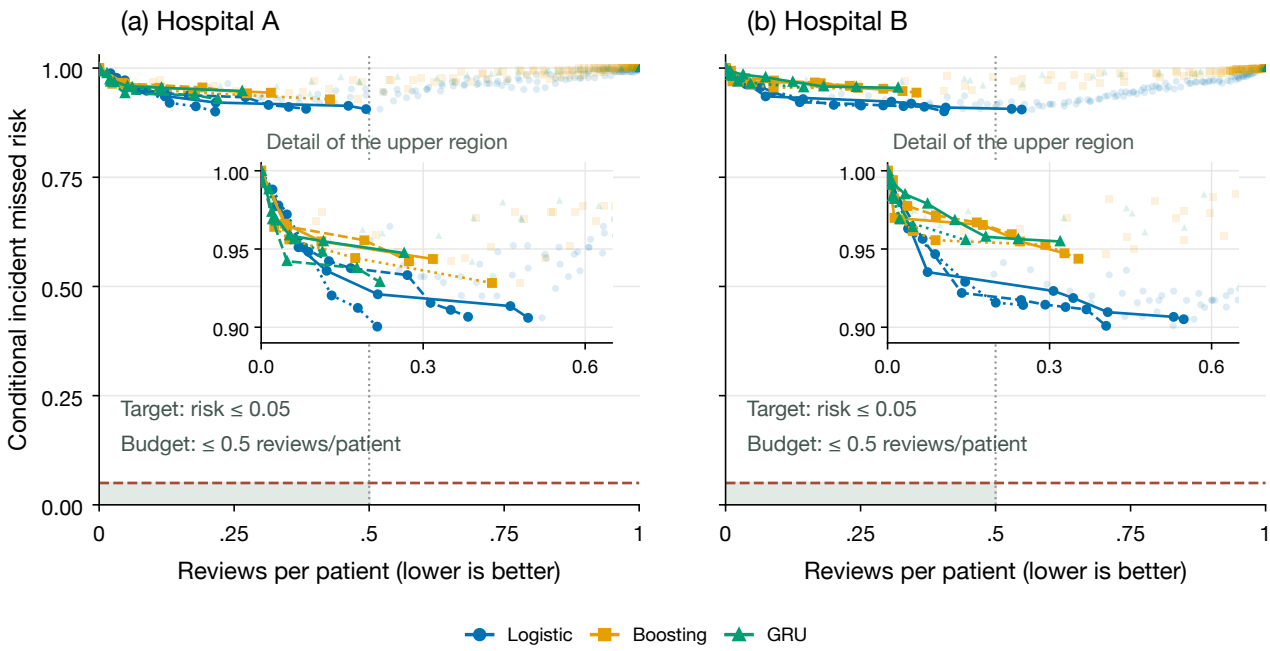


Figure 1: The tested first-alert grids remain far from the target region. All three seeds are shown (solid, dashed and dotted), with faint markers for dominated thresholds. Each model has 51 validation-quantile candidates plus endpoints before deduplication. Lines are frontier guides, not attainable interpolations. Shading marks risk ≤ 0.05 and workload ≤ 0.5. Panels show hospitals A and B; each inset enlarges risk 0.89–1.0 while the outer axes retain the complete risk range.

Table 3: Validation-selected operating policies, means over three seeds. "Repeated" reviews every above-threshold hour. Utility uses full-stay states; risk and review count use the retrospective pre-onset endpoint. No test metric chooses a threshold.

Model	Site	Policy	Missed risk	Reviews/patient	Utility
Logistic	A	First	0.936	0.215	0.333
Logistic	A	Repeated	0.298	8.077	0.342
Logistic	B	First	0.916	0.296	0.085
Logistic	B	Repeated	0.302	10.185	0.003
Boosting	A	First	0.960	0.193	0.332
Boosting	A	Repeated	0.318	6.741	0.380
Boosting	B	First	0.970	0.091	0.125
Boosting	B	Repeated	0.494	4.162	0.183
GRU	A	First	0.949	0.160	0.347
GRU	A	Repeated	0.406	6.124	0.352
GRU	B	First	0.971	0.081	0.155
GRU	B	Repeated	0.534	4.172	0.180

randomly assigned review nodes. This illustrates why review nodes alone do not enforce review, while remaining a synthetic structural test.

Figure 3 shows why fewer reviews cannot be interpreted in isolation. The GRU's review demand falls as more input groups are denied, while its already high missed risk increases. Logistic and boosting responses are less monotone because permission changes alter the score trajectories and the empirical fallback threshold. The released revocation analysis additionally separates admission opt-out from clearing state at hour 12. Neither intervention recovers a certified first-alert operating point.

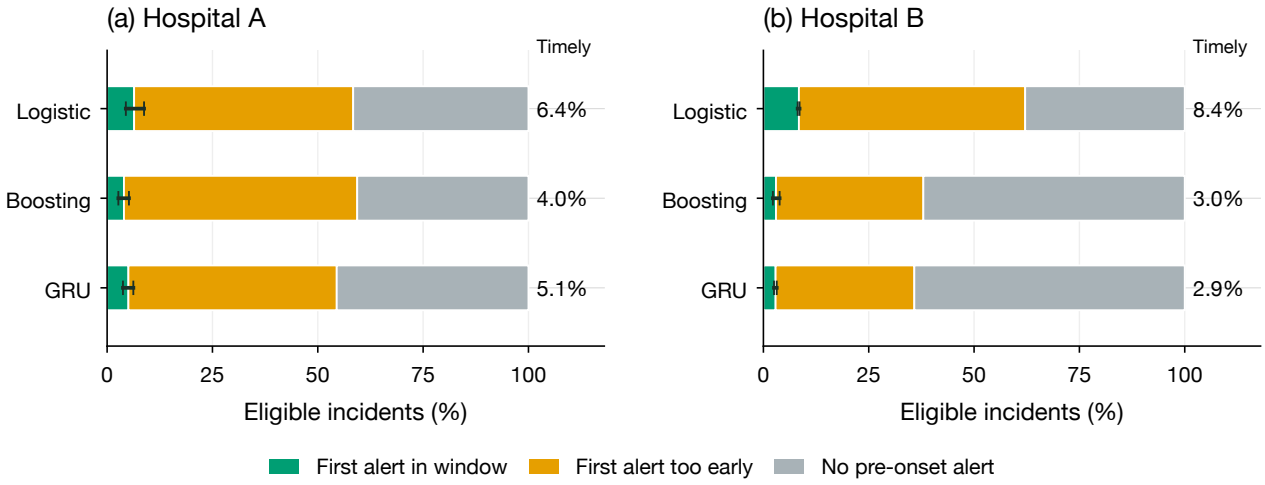


Figure 2: First-alert failures come from both early reviews and absent reviews. Bars show mean fractions across three seed-specific test sets at validation-utility-selected thresholds. Dark whiskers span the minimum and maximum timely fraction across seeds; they are descriptive ranges, not confidence intervals. Each label at the right gives the mean timely fraction.

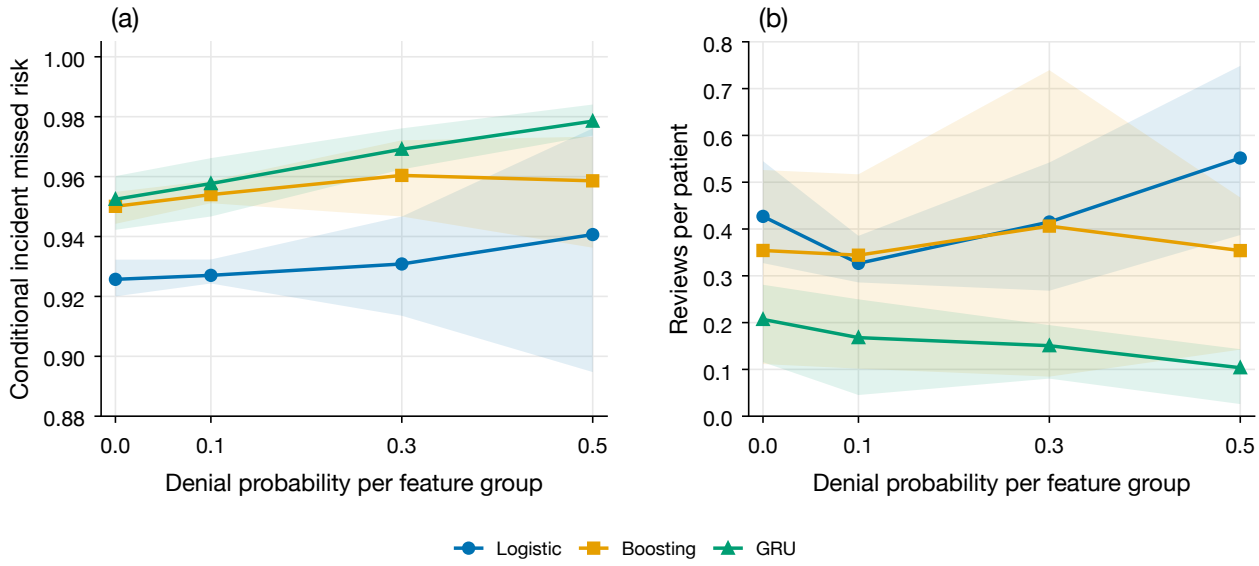


Figure 3: Admission opt-out changes incident risk and review demand together. Hospital A results use independently denied lab and vital groups and scenario-specific calibration-selected empirical fallback thresholds. Lines are three-seed means; shading spans the observed seed range. Panel (a) uses an expanded high-risk scale, not a zero baseline. No point carries a certified consent-scenario risk guarantee.

8 Discussion

8.1 What the study answers

The research question combines useful prediction, permission-respecting data use, controlled errors and effective human oversight. Our evidence supports different conclusions for these components. The graph representation makes specified permissions and review paths testable. The statistical construction gives a valid way to evaluate joint risk and workload on a frozen grid. The full-data experiment then produces a concrete negative answer for the first-alert family: the declared incident-risk target is not achieved, despite better discrimination and positive challenge utility. Actual clinician effectiveness and patient benefit were not measured.

This is more informative than observing that a bound failed to certify a weak model. The local confidence calculations exclude every tested candidate, and the test frontiers remain far above the 5% risk target. The calibration pools are large enough that a zero-miss candidate could pass the finite-sample correction. The principal obstacle in this experiment is the combination of score trajectories and the definition of timely first review. More calibration data alone

cannot turn a high population-risk candidate into a low-risk one.

The distinction also explains why the stronger predictors do not dominate the linear model on every endpoint. AUROC rewards correct ordering of individual hourly scores. Timely first review depends on the joint trajectory: scores before the window must remain below threshold, while at least one in-window score must cross it. The joint law of the pre-window and in-window maxima is therefore central. An improvement in hourly ranking need not improve that joint event, and a model that sends an early alert can receive utility credit while failing the strict first-review criterion.

8.2 How the alert policy changes the guarantee

Threshold nesting is often invoked to justify risk calibration, but the decision procedure sits between the score and the loss. Alert suppression, cooldown, a one-review limit or a persistent clinical assessment can change the loss family even when the score thresholds are unchanged. The counterexample in Proposition 1 isolates this issue without relying on a particular dataset or model architecture.

A useful operational policy may allow an early assessment to remain active, request a second review after deterioration, or suppress repeated notifications for a fixed period. None of these policies is ruled out by our results. Each would need a precise definition of review cost, the duration of an assessment and the event that counts as a successful action. The finite-grid method can evaluate non-monotone policies once their patient-level losses and workloads are specified, although a workload variable above one would require a bounded-mean or count-aware interval rather than the Bernoulli interval used here.

The distinction between infeasible and undetermined should remain visible in such evaluations. A confidence method that only returns accepted candidates cannot, by itself, distinguish an inadequate predictor from an underpowered calibration set. Our lower-bound exclusion criterion supplies that additional information, while limiting the conclusion to the actual finite grid. Extending an exclusion claim to all real-valued thresholds would require additional structure or a uniform result beyond the present analysis.

8.3 Consent as a changing input process

Permission changes alter more than the number of visible columns. A causal summary can retain earlier measurements; observation counts can reveal the measurement process; a recurrent state can encode information from a source that is no longer permitted. Clearing these paths is part of the implemented consent intervention. The conservative recurrent reset illustrates a real design tension: strict removal of mixed historical state can also remove useful permitted context.

The population consequences depend on who opts out, which sources are restricted and when permissions change. Our independent per-group denial process is a controlled stress test, not a model of real consent behavior. It is particularly important not to read the resulting cost curves as evidence that patients should disclose a particular type of information. Dynamic consent includes communication and participant agency [14]; predictive cost is only one input to that broader process.

The exchangeability statement clarifies the statistical boundary. A stable joint distribution of trajectories and permission states supports the certificate after the same frozen transformation is applied. Conditional outcome-ignorability of consent does not, by itself, make a changed deployment mechanism equivalent to the calibration mechanism. Hospital shift and consent shift can also occur together. Estimating a domain ratio from an initial feature vector cannot establish that the complete longitudinal patient, outcome and consent distribution has been correctly transported.

8.4 Oversight requires more than mandatory routing

The graph analysis and queue simulation answer complementary questions. The former asks whether a consequential action can be reached without traversing a review node. The latter asks whether a request receives a timely correct review under explicit assumptions. A path constraint can be satisfied even when every reviewer is overloaded or mistaken; a favorable queue result can coexist with an unmodeled bypass in a real software system.

The queue results are dominated by the upstream incident policy because a reviewer cannot repair a missing or mistimed request within the specified endpoint. This does not imply that reviewer quality is generally unimportant. It shows that improving one component cannot compensate for an event that the system never routes to that component in time. Clinical implementation evidence such as TREWS adoption [7] is needed to determine how review actually occurs, and externally validated alert-burden analyses such as Wong et al. [9] are needed to assess the workload that a hospital would experience.

9 Limitations and next evaluation

The warning window is deliberately strict and retrospective. An alert just before its start counts as a failure even if an earlier assessment would benefit a patient. Onset is reconstructed from the supplied label rather than independently adjudicated, and workload stops at a retrospectively defined time. The study consequently evaluates a well-defined benchmark requirement; it does not establish an optimal clinical timing rule. A prospective design should specify an observable review episode and account for the persistence of an assessment.

The model comparison is limited to three implementations and modest tuning budgets. It excludes dedicated

missingness-decay models, ensembles, richer temporal features and many possible training objectives. Class weighting affects probability calibration, and no post-hoc probability calibrator is fitted. The first-alert grid contains validation quantiles rather than every attainable threshold. Reported grid infeasibility should not be expanded into a claim about all models, all thresholds or all healthcare AI pipelines.

The sampling assumptions are also stronger than the available clinical record can verify. We treat benchmark patient files as independent units. Repeated patients across hospitals, temporal changes in care, correlated ward practices and label error could violate a deployment interpretation. Bootstrap intervals condition on fitted models and fixed splits; the three seeds reuse many patients. Group checks use a small set of recorded variables and do not establish broader fairness. The calibration-subset draws overlap, and no certified rule exists for a conditional test-risk exceedance distribution.

Consent and reviewers are simulated. There are no observed preference trajectories, clinician assessments, service-time measurements or prospective patient outcomes. Arrival and fatigue models are chosen sensitivity functions, not fitted behavioral laws. The graph is a model of information flow rather than an audit of a deployed EHR integration. Consent masks do not provide differential privacy, cybersecurity protection, deletion from previously trained model parameters or a complete account of institutional governance.

A next evaluation should first compare policies that permit continuing review or a documented cooldown, using the same incident definition and a workload unit that reflects the actual review episode. It should assess whether a feasible policy can be certified on local data and should prespecify how it responds when consent or hospital conditions change. Prospective human evaluation would then need to measure alert interpretation, correct action, response time, workload, overrides and patient outcomes. These are required measurements, not outcomes supplied by the present simulation.

10 Conclusion

A consent-aware clinical monitoring pipeline can be specified and evaluated with explicit data-flow, error and review-capacity requirements. In this public-data study, however, stronger prediction did not make the tested first-alert policies satisfy the joint requirement. The main lesson is methodological: certify the loss induced by the actual review policy, and distinguish useful scores, useful benchmark states and timely human action. The proposed evaluation makes that distinction measurable without treating structural compliance or simulated reviewers as evidence of clinical effectiveness.

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A Known-ratio transfer bound

Theorem 7 (Known-ratio shift certificate)**.** *Suppose the target patient law Q is absolutely continuous with respect to source law P, with known density ratio $0 \le w \le B_w$ and $\mathbb{E}_P w = 1$. For a fixed grid and n independent source patients define*

$$\widehat{a}_j = n^{-1} \sum_i w_i Z_i L_{ij}, \quad \widehat{b} = n^{-1} \sum_i w_i Z_i,$$
$$\widehat{c}_j = n^{-1} \sum_i w_i R_{ij}.$$

Set $\epsilon = B_w \sqrt{\log((2M+1)/\delta)/(2n)}$. With probability at least $1-\delta$, simultaneously for every j with $\widehat{b} > \epsilon$,

$$\mathbb{E}_Q[L_j \mid Z = 1] \le \min\left\{1, \frac{\widehat{a}_j + \epsilon}{\widehat{b} - \epsilon}\right\},$$
$$\mathbb{E}_Q[R_j] \le \min\{1, \widehat{c}_j + \epsilon\}.$$

Thus selecting a policy whose two upper bounds meet (α, K) controls both target quantities. If the denominator is nonpositive, the risk bound is uninformative.

This is a bounded-importance-weight Hoeffding argument. A feature-only density ratio is valid only under the corresponding conditional outcome invariance and support assumptions. A fitted, clipped domain classifier does not provide a known ratio or a verified bound on approximation error. Our empirical weighted comparison is explicitly exploratory, not an application of Theorem 7 with a guaranteed ratio.

B Proof of population feasibility

Proof of Theorem 2. A first alert is in the warning window exactly when no earlier score reaches threshold and some in-window score does. With inclusive alert ties, those events are $A < \tau$ and $B \ge \tau$. Thus $r(\tau) = 1 - H(\tau)$. Any review occurs exactly when $C \ge \tau$, giving $w(\tau) = W(\tau)$ by conditioning on Z. This proves the set characterization. The function W is non-increasing in τ, regardless of whether H is monotone. If the risk-feasible set has maximum $\tau_\star$, its workload is no larger than at every other risk-feasible threshold, proving Equation (8) and both directions of the budget equivalence. Timely success implies a review, so for every risk-feasible threshold $W(\tau) \ge \mathbb{P}(Z = 1, T_\tau \in [a, b]) = \pi H(\tau) \ge \pi(1-\alpha)$. If the risk-feasible set is empty, increasing workload cannot create a member of it. □

Proof of Corollary 3. With no possible earlier alert and $B = C$ on positives, $r(\tau) = \mathbb{P}(C < \tau \mid Z = 1) = F_1(\tau)$ by continuity. The unique maximal feasible threshold is q_α under the stated strictness and attainment assumptions. Substitution into Equation (7) gives $\pi\{1 - F_1(q_\alpha)\} + (1-\pi)\{1 - F_0(q_\alpha)\}$, and $F_1(q_\alpha) = \alpha$ completes the result. □

C Proof of finite-sample certification

Proof of Theorem 4. Condition on the fitted predictor and independently chosen candidate grid. For each policy, positive-patient losses are Bernoulli with mean r_j, conditional on the calibration positive count and membership; all-patient review indicators are Bernoulli with mean w_j. Inverting the binomial tails gives each one-sided interval coverage at least $1 - \eta$. This remains true unconditionally by averaging over the random positive count. Dependence between policies, or between risk and workload statistics, does not affect a union bound. There are $4M$ one-sided bounds, so all cover simultaneously with probability at least $1 - 4M\eta = 1 - \delta$. On this event, upper-bound feasibility implies true feasibility for every accepted policy, hence for any selection among them. If every policy has a lower bound violating at least one constraint, each policy violates that population constraint; therefore none on the grid is feasible. If neither condition is satisfied, the intervals do not resolve the question. The $[0, 1]$ convention at zero positives makes no unsupported claim. □

Proof of the group corollary. Apply the preceding coverage event within each prespecified group with error probability δ_g. A union bound over groups gives failure probability at most $\sum_g \delta_g$. Groups need not be independent. The conditional sampling assumptions must hold within each group, and post-hoc group discovery is not covered. □

D Proofs under consent and shift

Proof of Proposition 6. For measurable sets $B_1, \ldots, B_n$, independence of the input tuples gives $\mathbb{P}(\Phi(U_i, C_i, V_i) \in B_i,\ i = 1, \ldots, n) = \prod_i \mathbb{P}((U_i, C_i, V_i) \in \Phi^{-1}(B_i))$. Identical input laws and the same transformation give identical output laws. Conditioning independent patient tuples on their own positive-event indicators preserves the corresponding conditional product law. Nothing in the argument requires consent to be outcome-independent; it requires the complete joint patient/consent mechanism to be stable across calibration and future data. □

Proof of Theorem 7. The random variables wZL_j, wZ, wR_j all lie in $[0, B_w]$. One-sided Hoeffding bounds with failure probability $\delta/(2M+1)$ control the M numerator upper deviations, one denominator lower deviation, and M workload upper deviations. A union bound yields their simultaneous coverage. Change of measure gives $\mathbb{E}_P[wZL_j] = \mathbb{E}_Q[ZL_j]$, $\mathbb{E}_P[wZ] = \mathbb{P}_Q(Z = 1)$, and $\mathbb{E}_P[wR_j] = \mathbb{E}_Q[R_j]$. When the denominator lower bound is positive, dividing its numerator upper bound by that lower bound bounds the target conditional risk. Truncation at one uses the Bernoulli range. The same simultaneous event covers data-dependent selection among accepted candidates. □

E Seed-specific uncertainty and feasibility

E.1 Fixed-policy risk intervals.

The policy audit reports 36 validation-selected operating points with exact pointwise 95% binomial intervals in `selected_policies.csv`. They are not simultaneous across models, sites and seeds. The confidence intervals in that table assess realized test risk; the certificate uses an independent calibration population. No certified policy is available for a certified-policy test-risk distribution.

F Reproducibility and implementation checks

The public repository contains locked Python dependencies, exact data manifests, patient-level split assignments, fitted model outputs, frozen prediction arrays, tuning records and all result tables. The main full-release run used commit `fbfaff0d`; the frozen-policy audit used `652de9e8`. Exact source archives preserve both computations. The publication entry point runs a fresh full-data evaluation and then the policy audit, rather than requiring the original partial-run resume.

The main evaluation ran on an Apple M4 CPU with 16 GiB memory and two numerical-library/torch threads. The completed full-data run took 9 minutes 3 seconds. It reused a completed logistic seed-11 fit from a 2 minute 40 second partial run that failed during constant-feature binning for boosting. Hashes verify the reused outputs. The later policy audit took 1 minute 20 seconds and did not retrain predictors. Peak memory was not instrumented. Earlier exploratory studies are archived separately and are not included in these run times.

The corrected reviewer audit changes the false-action eligibility rule: a patient with any positive label is not treated as a true negative merely because the patient is ineligible for the strict incident endpoint. The original queue output is retained for provenance and labeled superseded. Predictor scores, incident-loss calculations and calibration-certificate outcomes are unchanged by that correction.

Twenty-three implementation checks pass. These include causal feature invariance, consent/purpose masks, cache revocation, review-path equivalence, the first-alert non-monotonicity counterexample, exact binomial boundary cases, the distinction between infeasibility and insufficient evidence, agreement with official utility, handling of constant/all-missing features and false-action eligibility. These checks are mechanised evidence about the implementation. The proofs in the appendix establish the mathematical results independently of those tests.

The primary public data retain their CC BY 4.0 license and the official evaluator retains its BSD license. No access-controlled dataset agreement is accepted by the code. No new patient or clinician recruitment occurred. This draft does not assert a project-specific institutional approval or exemption; applicable institutional requirements should be resolved before submission or any prospective study.

AI assistance was used in implementation, mathematical drafting, analysis and manuscript preparation. No LLM is used as a predictor or reviewer in the evaluated pipeline. Author responsibility includes checking the mathematical arguments, source attributions and interpretation before submission. The code, data, paper source and interactive results are available at `https://github.com/aravinds-kannappan/Clinical-AI` and `https://consent-clinical-ai.vercel.app`.

Table 4: All-seed pre-onset AUROC with 95% percentile intervals from 500 whole-patient bootstrap replicates. Brier scores are point estimates; their intervals are released in the machine-readable metrics table. Intervals describe patient-sampling uncertainty conditional on each fitted model.

Model	Seed	AUROC A [95% CI]	AUROC B [95% CI]	Brier A	Brier B
Logistic	11	0.750 [0.724, 0.776]	0.651 [0.634, 0.669]	0.015	0.014
Logistic	29	0.738 [0.711, 0.766]	0.679 [0.661, 0.695]	0.187	0.235
Logistic	47	0.746 [0.718, 0.773]	0.667 [0.649, 0.682]	0.189	0.210
Boosting	11	0.792 [0.769, 0.818]	0.713 [0.696, 0.732]	0.098	0.060
Boosting	29	0.799 [0.775, 0.821]	0.731 [0.715, 0.746]	0.092	0.055
Boosting	47	0.792 [0.767, 0.816]	0.748 [0.730, 0.763]	0.014	0.009
GRU	11	0.789 [0.766, 0.816]	0.742 [0.726, 0.759]	0.056	0.043
GRU	29	0.787 [0.761, 0.813]	0.752 [0.737, 0.766]	0.057	0.036
GRU	47	0.756 [0.732, 0.782]	0.763 [0.749, 0.777]	0.066	0.053

Table 5: Feasibility of the validation-selected first-alert policy under conditional incident loss. Each row aggregates three seeds: "Y" means every seed meets the empirical criterion; "N" means none does. Storage and review checks concern the specified graph, not a deployed system. Actual human oversight was not measured.

Model	Site	Utility > 0	Risk $\leq .05$	Reviews $\leq .5$	Storage cut	Review path
Logistic	A	Y	N	Y	Modeled	Modeled
Logistic	B	Y	N	Y	Modeled	Modeled
Boosting	A	Y	N	Y	Modeled	Modeled
Boosting	B	Y	N	Y	Modeled	Modeled
GRU	A	Y	N	Y	Modeled	Modeled
GRU	B	Y	N	Y	Modeled	Modeled